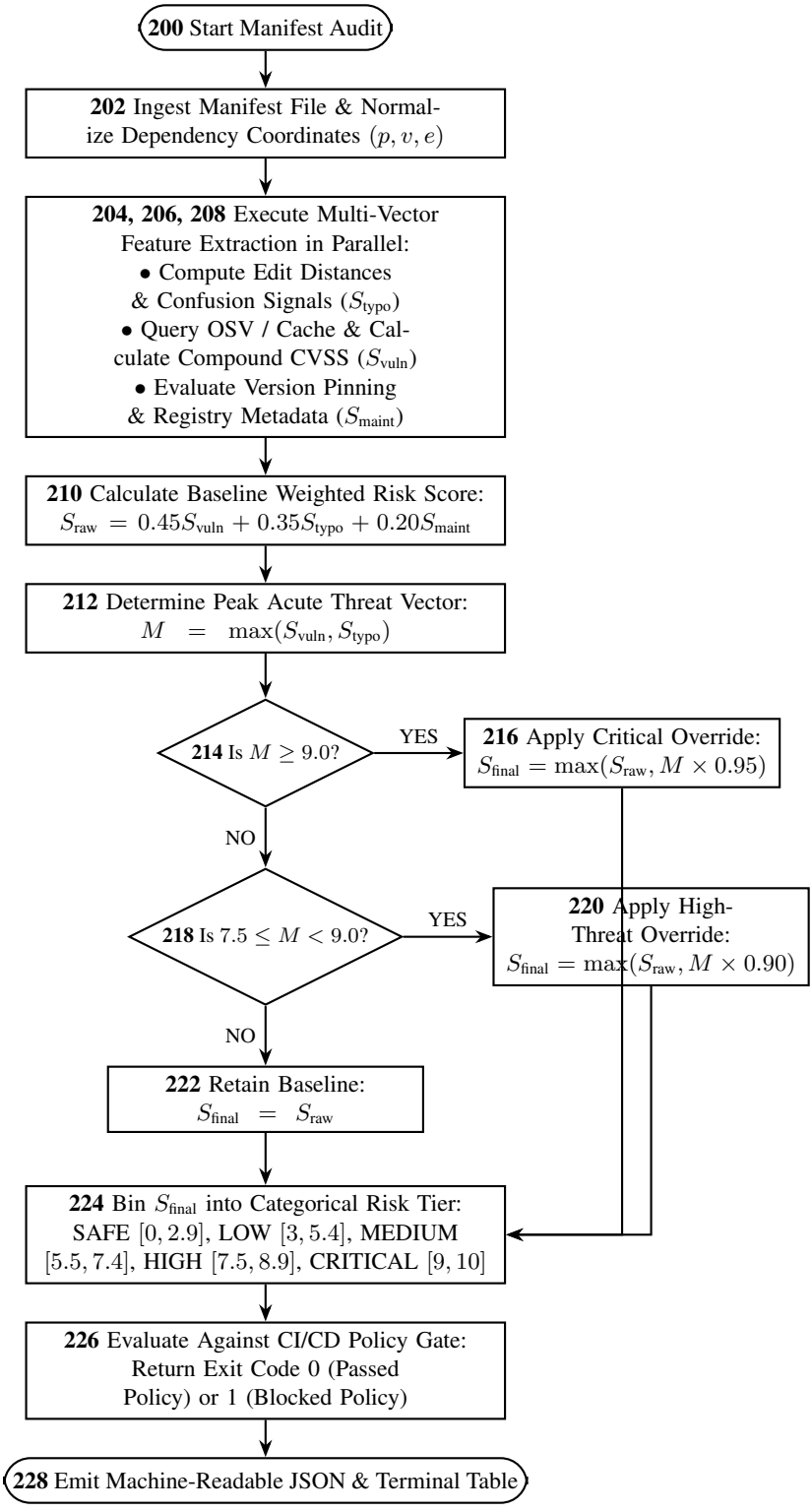

200 Start Manifest Audit
202 Ingest Manifest File & Normalize Dependency Coordinates (p, v, e)
204, 206, 208 Execute Multi-Vector Feature Extraction in Parallel:
• Compute Edit Distances & Confusion Signals (S_{typo})
• Query OSV / Cache & Calculate Compound CVSS (S_{vuln})
• Evaluate Version Pinning & Registry Metadata (S_{maint})
210 Calculate Baseline Weighted Risk Score:
$S_{\text{raw}} = 0.45S_{\text{vuln}} + 0.35S_{\text{typo}} + 0.20S_{\text{maint}}$
212 Determine Peak Acute Threat Vector:
$M = \max(S_{\text{vuln}}, S_{\text{typo}})$
214 Is $M \geq 9.0$?
YES
216 Apply Critical Override:
$S_{\text{final}} = \max(S_{\text{raw}}, M \times 0.95)$
NO
218 Is $7.5 \leq M < 9.0$?
YES
220 Apply High-Threat Override:
$S_{\text{final}} = \max(S_{\text{raw}}, M \times 0.90)$
NO
222 Retain Baseline:
$S_{\text{final}} = S_{\text{raw}}$
224 Bin S_{final} into Categorical Risk Tier:
SAFE $[0, 2.9]$, LOW $[3, 5.4]$, MEDIUM $[5.5, 7.4]$, HIGH $[7.5, 8.9]$, CRITICAL $[9, 10]$
226 Evaluate Against CI/CD Policy Gate:
Return Exit Code 0 (Passed Policy) or 1 (Blocked Policy)
228 Emit Machine-Readable JSON & Terminal Table